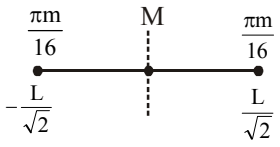
SOLUTION
SECTION-I

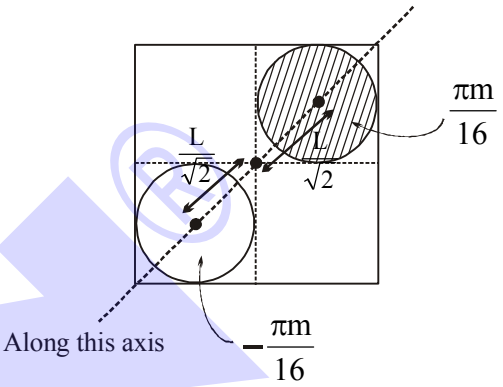
Single Correct Answer Type **4 Q. [3 M (–1)]**

1. Ans. (D)

Sol.



$$X_{Cr} = \frac{M\pi\left(\frac{L}{2}\right)^2}{4L^2} \times \left(L\hat{i} + L\hat{j}\right) \Big/ M$$



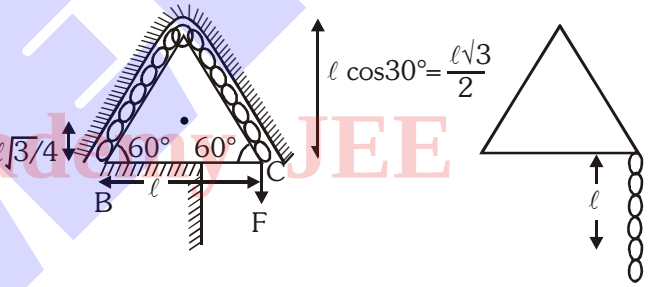
2. Ans. (D)

Sol. Change in position of CM in vertical direction

$$(\Delta h) = \left(\ell\frac{\cos 30°}{2}\right) + \ell$$

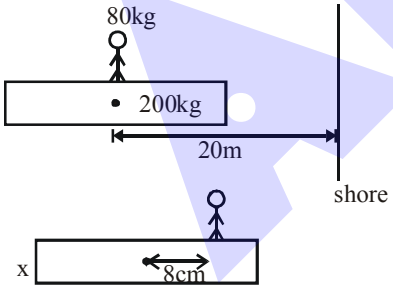
$\therefore$ Work done by gravity

$$= + mg\Delta h = + mg\left(\frac{\ell\sqrt{3}}{4} + \ell\right) = + mg\ell\left(\frac{\sqrt{3}+4}{4}\right)$$



3. Ans. (C)

Sol. Let the displacement of boat be x towards the shore



$$80\,(x + 8) + 200\,x = 0$$

$$\Rightarrow x = \frac{-640}{280} = -\frac{16}{7}$$

$$\therefore \quad (\Delta x)_{mon} = 8 - \frac{16}{7} = \frac{40}{7}$$

$\therefore$ New distance from shone

$$= 20 - \frac{40}{7} = \frac{100}{7} = 14.28\text{m} \approx 14.3\text{ m}$$

Rankers Academy JEE

For More Material Join: @JEEAdvanced_2024

4. Ans. (C)

Multiple Correct Answer Type **6 Q. [4 M (–1)]**

5. Ans. (B, D)

Sol. $W_{man} = \Delta KE$

From momentum conservation $mv_{man} = Mv$

For (A) : Work done by man on boat = $\Delta KE_{boat} = \frac{1}{2}Mv^2$

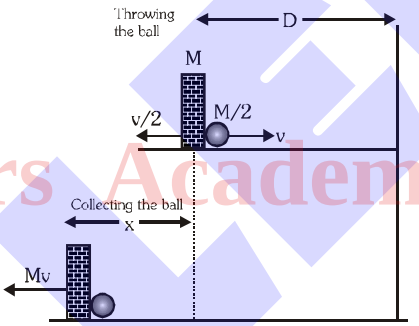
For (B) : $W_{man\ total} = \Delta KE_{system} = \frac{1}{2}m\frac{M^2v^2}{m^2} + \frac{1}{2}Mv^2$

For (C) : $V_{CM} = 0$ {momentum conservation}

For (D) : $W_{man\ on\ himself} = \Delta KE_{man} = \frac{1}{2}\frac{M^2}{m}v^2$

6. Ans. (A,D)

Sol. From conservation of linear momentum velocity of man = $\frac{v}{2}$



From conservation of linear momentum, momentum of man + ball system after he collects the ball

$p_2 = \frac{Mv}{2} + \frac{M}{2}v = Mv$ ...(i)

For (A) : Velocity of man + ball system

$= \frac{\text{momentum}}{\text{mass}} = \frac{Mv}{M + \frac{M}{2}} = \frac{2v}{3}$

For (B) : Linear impulse by the ball on man
= change in momentum of the man

$= M\left(\frac{2v}{3}\right) - \frac{Mv}{2} = \frac{Mv}{6}$

For (D) : $t = \frac{x}{v/2} = \frac{2D + x}{v} \Rightarrow x = 2D$.

So time $t = \frac{x}{v/2} = \frac{4D}{v}$

7. Ans. (C,D)

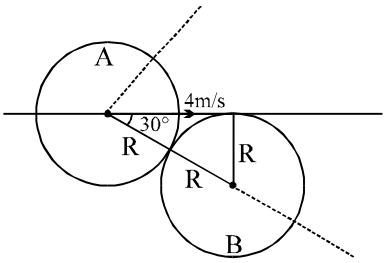
Sol. Linear momentum conservation along horizontal can be made only if there is no friction or non–impulsive friction.

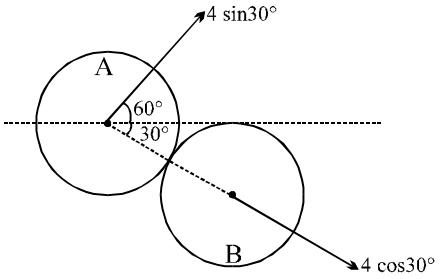
For More Material Join: @JEEAdvanced_2024

8. Ans. (A, C)

9. Ans. (A,C)

Sol. Before collision **After collision**





$v_A = 4 \sin 30° \, [\cos 60 \hat{i} + \sin 60 \hat{j}]; \; v_A = \hat{i} + \sqrt{3}\hat{j}$

$\overrightarrow{J_{A \text{ on } B}} = m\overrightarrow{V_{B_f}} - \overrightarrow{V_{B_i}} = m\,[4\cos 30°(\cos 30° \hat{i} - \sin 30° \hat{j}) - 0] = (3mi - \sqrt{3}mj)$ kg-m/s

(D) There is a net loss in the kinetic energy of the two particle system in the collision.

10. Ans. (B, C)

Sol. $h' = \dfrac{e^2 v^2 \sin^2 \alpha}{2g} = e^2 \dfrac{2gh}{2g} \times \dfrac{3}{4}$

$h' = \dfrac{3h}{4} e^2$

Linked Comprehension Type (2 Para × 3Q.) [3 M (-1)]
(Single Correct Answer Type)

11. Ans. (A)

Sol.



Just before collision Just after collision

$v'_1 = (1+e)v_c - ev_1 = 2\left(\dfrac{Mv_0}{M+m_1}\right) - v_0$

$v'_2 = (1+e)v_c - ev_2 = 2\left(\dfrac{Mv_0}{M+m_1}\right)$

Energy gain of quark m_1 is

$= \dfrac{1}{2} m_1 (v'_2)^2$

$= \dfrac{1}{2} m_1 \left[\dfrac{2Mv_0}{M+m_1}\right]^2$

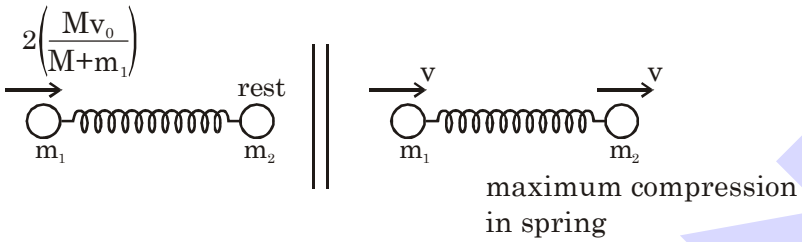
For More Material Join: @JEEAdvanced_2024

$$= 4\left[\frac{Mm_1}{(M+m_1)^2}\right]\left(\frac{1}{2}Mv_0\right)^2$$

$$= 4\left[\frac{Mm_1}{(M+m_1)^2}\right]E$$

12. Ans. (A)

Sol.



Momentum conservation

$$(m_1+m_2)v = m_1 \cdot 2\left(\frac{Mv_0}{M+m_1}\right)$$

$$v = \frac{2m_1Mv_0}{(M+m_1)(m_1+m_2)}$$

$$(KE)_{min} = \frac{1}{2}(m_1+m_2)v^2$$

$$= 4\left[\frac{Mm_1^2}{(m_1+m_2)(M+m_1)^2}\right]\left(\frac{1}{2}Mv_0^2\right)$$

$$= \frac{4Mm_1^2 \cdot E}{(M+m_1)^2(m_1+m_2)}$$

13. Ans. (A)

Sol. Internal energy of meson

$$= \frac{1}{2}\mu \cdot v_{rel}^2$$

$$= \frac{1}{2}\left(\frac{m_1m_2}{m_1+m_2}\right)\left(\frac{2Mv_0}{M+m_1}\right)^2$$

Bankers Academy JEE

For More Material Join: @JEEAdvanced_2024

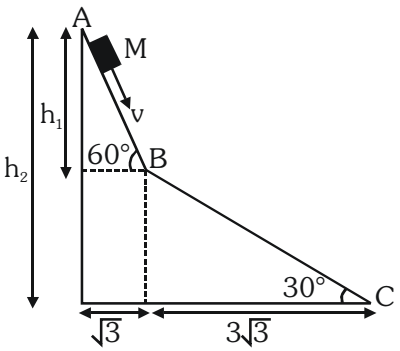
14. Ans. (B)

Sol. $\frac{h_1}{\sqrt{3}} = \tan 60^\circ \Rightarrow h_1 = 3m$

$\frac{h_2 - h_1}{3\sqrt{3}} = \tan 30^\circ$

$\Rightarrow h_2 - h_1 = 3$

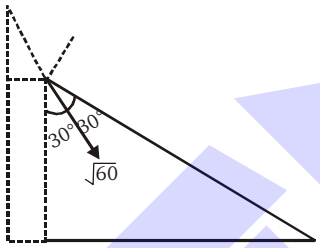
$\Rightarrow h_2 = 6m$



Velocity of block just before collision at B

$= \sqrt{2gh} = \sqrt{2 \times 10 \times 3} = \sqrt{60}$ ms^{-1}

For totally inelastic collision velocity of block along normal to BC becomes zero and since there is no impulse along BC so momentum (velocity) along BC remains unchanged. speed of block just after collision



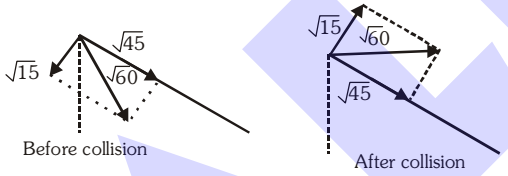
$v_B = \sqrt{60}\cos 30^\circ = \sqrt{60} \times \frac{\sqrt{3}}{2} = \sqrt{45}$ms^{-1}

15. Ans. (B)

Sol. $v_c^2 = v_B^2 + 2g(h_2 - h_1) \Rightarrow v_c = \sqrt{45 + 2 \times 10 \times 3} = \sqrt{105}$ ms^{-1}

16. Ans. (C)

Sol.



vertical component of velocity = $\sqrt{15}\sin 60^\circ - \sqrt{45}\sin 30^\circ = 0$

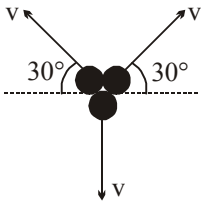
SECTION-II

Numerical Answer Type Question **2Q.[3M (0)]**

(upto second decimal place)

17. Ans. 50

Sol. $v_1 = v_2 = v_3$



For More Material Join: @JEEAdvanced_2024

$$h = vt + \frac{1}{2}gt^2 = v \cdot 2 + \frac{1}{2} \cdot g(2)^2$$

$$h = -\frac{v}{2} \cdot t_1 + \frac{1}{2}\, gt_1^{\,2} = -\frac{v}{2} \cdot 4 + \frac{1}{2} \cdot g \cdot 4^2$$

18. Ans. 10

Sol. $50\,(L - x) - 150\,x = 0 \Rightarrow 50\,L - 200\,x = 0 \Rightarrow x = \frac{L}{4}$

Similarly in y direction : $y = \frac{L}{4}$

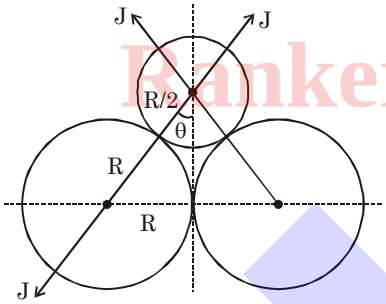
$$S = \sqrt{x^2 + y^2} \Rightarrow S = \frac{L}{2\sqrt{2}} = \frac{20\sqrt{2}}{2\sqrt{2}} = 10$$

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) 2 Q. [4 M (0)]

19. Ans. 1

Sol.



J = impulse during collision between a larger ball and smaller ball

For smaller ball

$2J\cdot\cos\theta = mv_0$ ... (i)

For larger ball, $J \sin\theta = mv$ ... (ii)

From (i) & (ii)

$$v = \frac{v_0}{2}\tan\theta$$

$$= \frac{\sqrt{5}}{2}\left[\frac{R}{\sqrt{\left(\frac{3R}{2}\right)^2 - R^2}}\right]$$

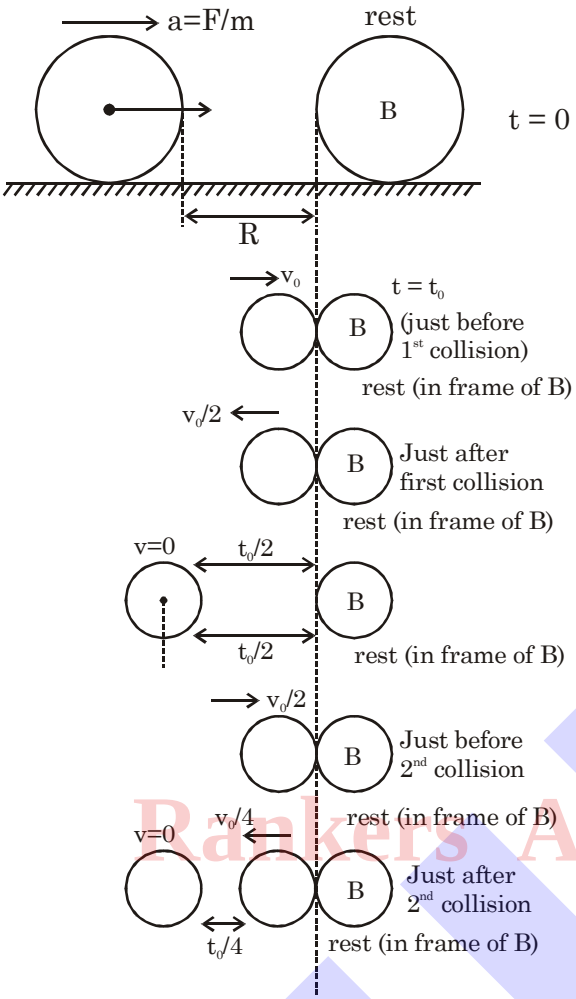
$$= \left(\frac{\sqrt{5}}{2}\right)\left(\frac{2}{\sqrt{5}}\right)$$

$= 1$ m/s

For More Material Join: @JEEAdvanced_2024

20. Ans. 2

Sol.



$$v_0 = \sqrt{2aR}$$

$$v_0 = at_0$$

$$t_0 = \frac{v_0}{a} = \sqrt{\frac{2R}{a}}$$

$$\Delta t = t_0 + 2\left[\frac{t_0}{2} + \frac{t_0}{4} + \frac{t_0}{8} + ...\infty\right] - t_0$$

$$= 2\left[\frac{t_0}{2} + \frac{t_0}{4} + \frac{t_0}{8} + ...\infty\right]$$

$$= 2 \cdot t_0\left[\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + ...\infty\right]$$

$$= \frac{1}{2} \cdot 2\sqrt{\frac{2R}{a}}\left[1 + \frac{1}{2} + \frac{1}{2^2} + ...\infty\right]$$

Rankers Academy JEE

For More Material Join: @JEEAdvanced_2024

$$= \sqrt{\frac{2R}{F/m}}\left[\frac{1}{1-\frac{1}{2}}\right]$$

$$= 2\sqrt{\frac{2R}{F/m}}$$

$$= 2\sqrt{\frac{2\times 1}{100/50}}$$

$= 2$

SECTION-IV

Matrix Match Type (4 × 5) **3 Q. [8 M (for each entry +2(0)]**

21. Ans. (A) PQRS (B) PQRS (C) PQRS (D) QS

Sol. **For (A) :** $F_{net_{external}} = 0$ $\Rightarrow a_{CM} = 0$
$\Rightarrow$ momentum is conserved
$\Rightarrow$ work done by external=0 $\Rightarrow$ Mechanical energy is conserved

Potential energy gets converted to K.E. $\Rightarrow$ KE $\neq$ constant

For (B) : $F_{net_{external}} = 0$ $\Rightarrow a_{CM} = 0$
$\Rightarrow$ momentum is conserved
$\Rightarrow$ work done by external=0 $\Rightarrow$ Mechanical energy is conserved

KE gets stored in the masses during collision as PE $\Rightarrow$ KE $\neq$ constant

For (C) : $F_{net_{external}} = 0$ $\Rightarrow a_{CM} = 0$
$\Rightarrow$ momentum is conserved
$\Rightarrow$ work done by external=0 $\Rightarrow$ Mechanical energy is conserved

KE gets stored in the masses during collision as PE $\Rightarrow$ KE $\neq$ constant

For (D) : $F_{net_{external}} \neq 0$ $\Rightarrow a_{CM} \neq 0$
$\Rightarrow$ momentum of system is not conserved

K.E. gets conserved in PE of spring $\Rightarrow$ KE $\neq$ constant.

22. Ans. (A)→(P,R,T); (B)→(Q,S); (C)→(P,S,T); (D)→(P, S, T)

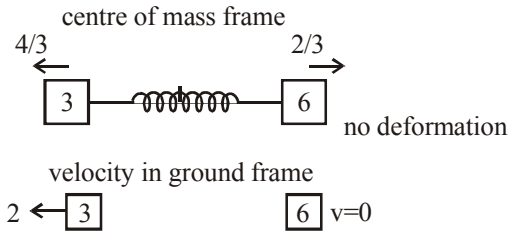
Sol. No external force along horizontal in A, C, D so center of mass will remain at rest and linear momentum will always remain constant.
In B work is done by F so mechanical energy increases.
In C, D internal forces of man does work so mechanical energy increases

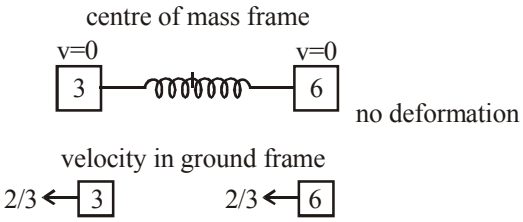
For More Material Join: @JEEAdvanced_2024

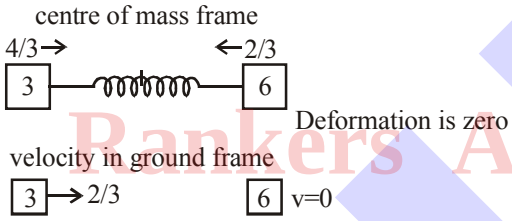
23. **Ans. (A)-P,Q,R,S; (B)-P,R,S; (C)-P,Q; (D)-P,Q**

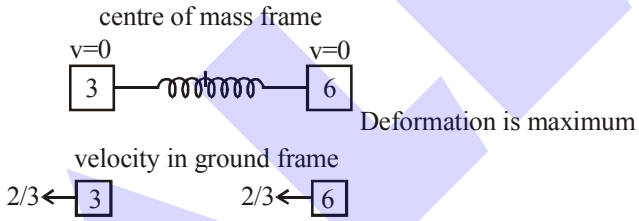
Sol. 

$$v_{CM} = \frac{3\times 2}{9} = \frac{2}{3}$$

(A)


(B)


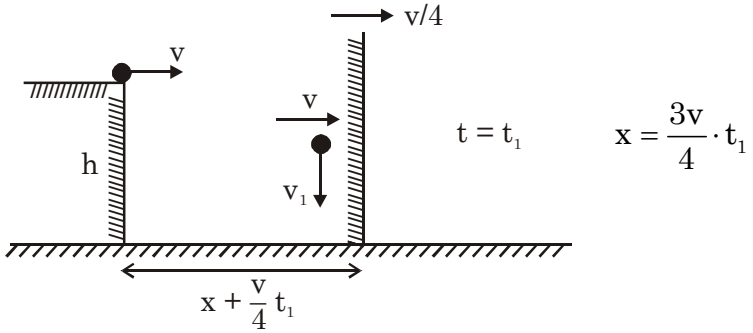
(C)


(D)


Subjective Type **7 Q. [4 M (0)]**

24. **Ans.** $T = 2\sqrt{\frac{2h}{g}}$, $x = v\sqrt{\frac{h}{2g}}$

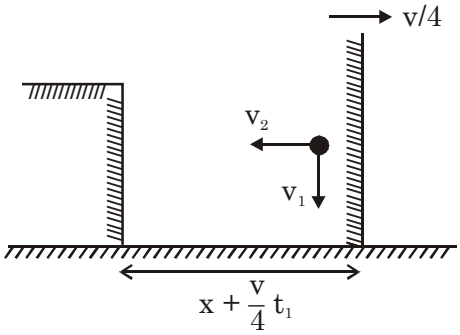
Sol. Just before collision with wall



$t = t_1$ $\quad x = \frac{3v}{4}\cdot t_1$

For More Material Join: @JEEAdvanced_2024

Just after collision with wall



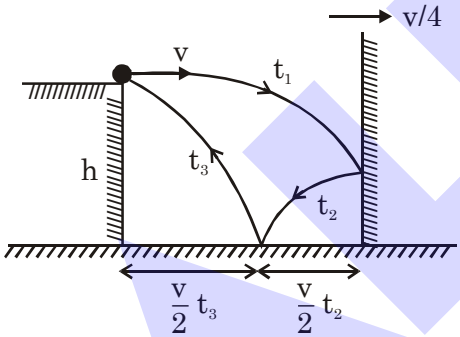
$$e = 1 = \frac{v_2 + v/4}{v - v/4}$$

$$\Rightarrow v_2 = \frac{v}{2}$$

Due to collision of particle with wall, there is no effect on vertical motion of particle.

$\therefore$ Total time $T = 2\sqrt{\frac{2h}{g}}$

Rankers Academy JEE



$$t_3 = \sqrt{\frac{2h}{g}}$$

$$t_1 + t_2 = \sqrt{\frac{2h}{g}} = t_3$$

$$\frac{v}{2}\cdot t_3 + \frac{v}{2}\cdot t_2 = x + \frac{v}{4}\cdot t_1$$

$$\frac{v}{2}\cdot t_3 + \frac{v}{2}(t_3 - t_1) = x + \frac{v}{4}\cdot t_1$$

For More Material Join: @JEEAdvanced_2024

$$v \cdot t_3 - \frac{v}{2} \cdot t_1 = x + \frac{v}{4} \cdot t_1$$

$$vt_3 = x + \frac{3}{4} \cdot v \cdot t_1$$

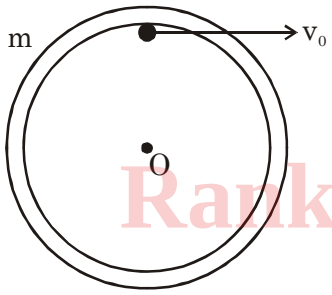
$$vt_3 = 2x$$

$$x = \frac{v}{2} \cdot t_3 = \frac{v}{2}\sqrt{\frac{2h}{g}}$$

$$x = v\sqrt{\frac{h}{2g}}$$

25. **Ans.** Straight line, $\frac{2\pi r}{v_0}$, $\frac{\sqrt{3}v_0}{2}$

Sol.



For given system (ring + particle) $\Rightarrow F_{ext} = 0$

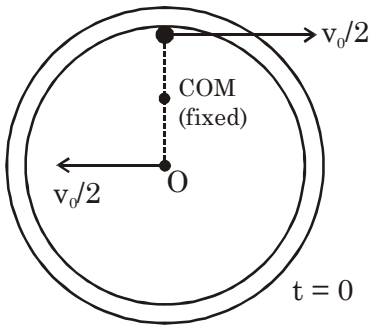
$\vec{v}_{cm}$ = constant

$$= \frac{mv_0}{2m}$$

$$= \frac{v_0}{2}$$

$\therefore$ Path of COM is straight line

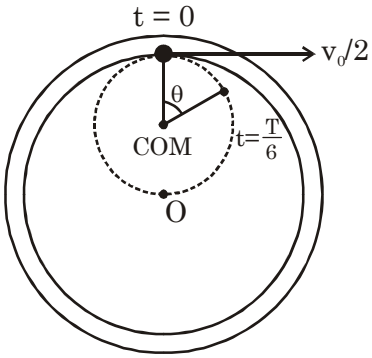
In COM frame $\Rightarrow$



For More Material Join: @JEEAdvanced_2024

In COM frame particle moves in a circle with constant speed $\frac{v_0}{2}$. Velocity of tyre again becomes zero after one complete revolution.

$$T = \frac{2\pi r/2}{v_0/2} = \frac{2\pi r}{v_0}$$



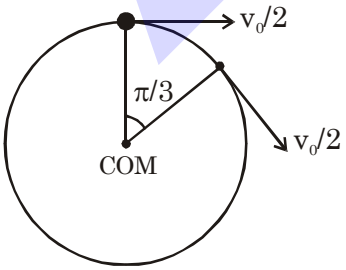
$$\text{At } t = \frac{T}{6}$$

$$\theta = \omega \cdot \frac{T}{6}$$

$$= \frac{v_0}{2 \cdot \frac{r}{2}} \cdot \frac{T}{6}$$

$$= \frac{v_0}{6r} \cdot \frac{2\pi r}{v_0}$$

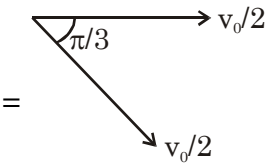
$$= \frac{\pi}{3}$$



$$\vec{v}_{Particle} = \vec{v}_{Particle,COM} + \vec{v}_{COM}$$

Rankers Academy JEE

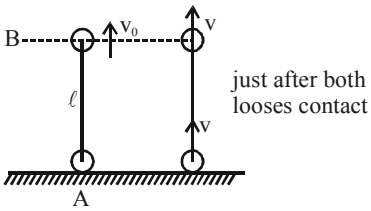
For More Material Join: @JEEAdvanced_2024



$$\vec{v}_{Particle} = \frac{v_0}{2}\sqrt{3}$$

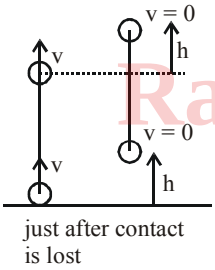
26. **Ans.** ℓ

Sol.



$$v_0^2 = 6\,g\ell - 2g\ell \Rightarrow v_0 = \sqrt{4g\ell}$$

$$2mv = mv_0 \Rightarrow v = \frac{v_0}{2} = \sqrt{g\ell}$$



just after both looses contact $\Rightarrow v_{cm} = \frac{mv + mv}{2m} = v = \sqrt{g\ell}$ & $a_{cm} = g$

Now just after both looses contact

$\Rightarrow 0^2 = v_{cm}^2 - 2gh$

$\Rightarrow h = \frac{\ell}{2}$ = displacement of each block after both looses contact.

Displacement of COM till only one block moves upward is

$\Rightarrow (\Delta x_{cm})_1 = \frac{m\ell}{2m} = \frac{\ell}{2}$

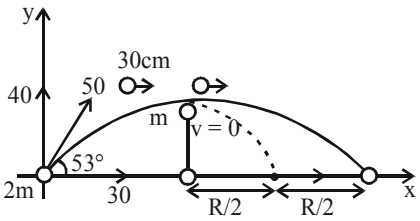
$\therefore$ Total upward displacement of COM is $= \frac{\ell}{2} + \frac{\ell}{2} = \ell$

Rankers Academy JEE

For More Material Join: @JEEAdvanced_2024

27. **Ans.** (i) 360 m, (ii) 10800 J

Sol.



(i) Range $R = \frac{50^2 \times 2\sin 53°.\cos 53°}{10} = 250 \times 2 \times .6 \times .8$

$R = 240$ m

$\therefore d = R + \frac{R}{2} = \frac{3R}{2} = 360$ m

(ii) From $P_f = P_i$ in horizontal

$2m \times 30 = mv$

$\Rightarrow v = 60$ m/s

Just before explosion

$K_i = \frac{1}{2}.2m.v_x^2$ $\qquad 2m = 24 \Rightarrow m = 12$

$= \frac{1}{2} \times 24 \times (30)^2 = \frac{1}{2} \times 21600J = 10800J$

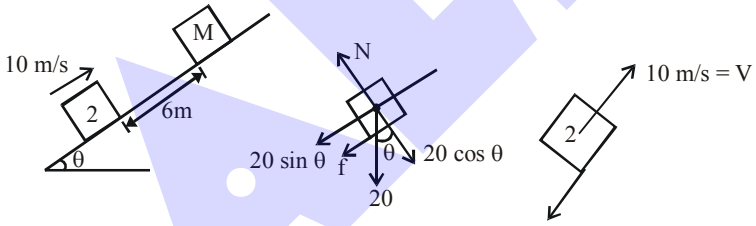
Just after explosion

$k_f = \frac{1}{2}mv^2 = \frac{1}{2} \times 12 \times 60^2 = 21600J$

$\therefore$ Energy released $= k_f - k_i = 21600 - 10800 = 10800$ J

28. **Ans.** $e = (5 + \sqrt{3})/8$, $M = 26/\sqrt{3}$ kg

Sol.



During upward motion of 2 kg

$f = \mu . 20 \cos\theta = .25 \times 20 \cos\theta$

$f = 5$

$\therefore a_{2kg} = \frac{20\sin\theta + 5}{2} = \frac{20 \times 0.05 + 5}{2} = 3$ m/s^2

velocity of 2kg just before collision is

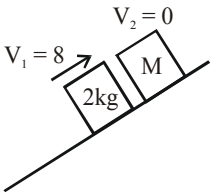
$V_1 = \sqrt{10^2 - 2 \times 3 \times 6} = \sqrt{64}$

$V_1 = 8$ m/s

$\therefore$ Condition of just before collision

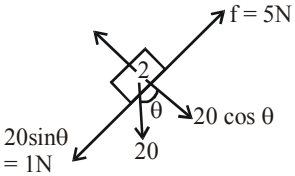
Rankers Academy JEE

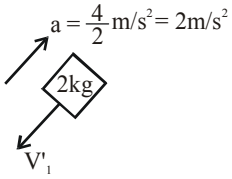
For More Material Join: @JEEAdvanced_2024



Just after collision

During downward motion of 2kg



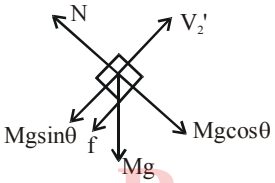


During motion of 2kg from P to initial position

$\Rightarrow 1^2 = V_1'^2 - 2 \times 2 \times 6$

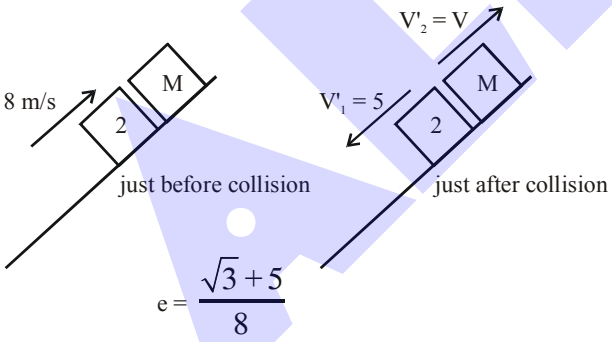
$\Rightarrow V_1'^2 = 25 \Rightarrow V_1' = 5\text{m/s}$

Upward motion of M

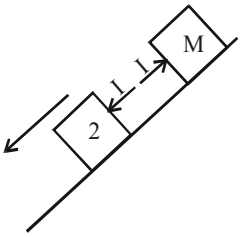


$$a = \frac{Mg\sin\theta + \mu Mg\cos\theta}{M} = g(\sin\theta + \mu\cos\theta) = 10(.05 + .25 \times 1) = 3\text{m/s}^2$$

Now $0^2 = V_2'^2 - 2 \times 3 \times 0.5 \Rightarrow V_2' = \sqrt{3}\text{m/s}$



To calculate (M =)



During collision

Rankers Academy JEE

For More Material Join: @JEEAdvanced_2024

Impulse on 2 kg is
$= (\Delta P)_{2kg}$
$= 2[5 - (-8)] = 26$ N.S.
impulse on Mkg is = 26

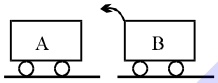
$M \cdot \sqrt{3} = 26$

$\Rightarrow M = \frac{26}{\sqrt{3}} kg$

29. Ans. $v_B = 55/36$m/s, $v_A = 11/6$ m/s, solved

Sol. Given $v_{D/C} = 6$ m/sec
Case Ist
C.O.L.M.

$O = M_D \vec{v}_D + m_B \vec{v}_B$



$\vec{v}_D = -6i + \vec{v}_B$

$\therefore$ i.e. $v_D = -5\hat{i}$ toword A

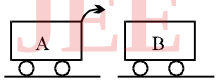
Case II[nd] In this condition when dog & trolly A will move with common speed again
C.O.L.M. $-5i \times 4 = (20 + 4)\, v_{common}$

$v_{common} = -5/6\,\hat{i}$

so will jumping from A $\rightarrow$ B

$\vec{v}_D = 6\hat{i} + \vec{v}_A$

$-(24) \times \frac{5}{6}\hat{i} = \left(\left(6\hat{i} + \vec{v}_A\right) \times 4 + \vec{v}_A \cdot 20\right)$



$v_A = -\frac{11}{6}\hat{i}$

Case III[rd]

$v_D = 6 - \frac{11}{6}\hat{i} = \frac{25}{6}\hat{i}$

$M_D \vec{v}_D + M_B \vec{v}_B = (M_B + M_D)\, v_{common}$

$\therefore$ $v_{common} = \frac{11}{6}$ m/sec Ans.

30. Ans. (a) $(ku - g)e^{-bt}$ (b) $[\{(ku - g)/b\}(1 - e^{-bt})]$, (c) $(ku - g)/b$

Rankers Academy JEE

For More Material Join: @JEEAdvanced_2024